

GOUTHAM SHIMOGA UMESH

Tempe, AZ | (602) 348-6090 | gouthamshimogaumesh@gmail.com | [LinkedIn](#) | [Portfolio](#) | F-1 OPT (Authorized to Work in US)

TECHNICAL SKILLS

CAD & Design: CATIA V5, SolidWorks, Creo Parametric, AutoCAD, ANSYS DesignModeler, 3D Modeling, Engineering Drawings, GD&T
Analysis: ANSYS Fluent (CFD), ANSYS Static Structural (FEA), Thermal Analysis, Heat Transfer, Multiphase Flow, Hand Calculations
Manufacturing & Tools: DFM/DFA, 3D Printing, Assembly Processes, Root Cause Analysis, MATLAB, Python, SAP ERP, MS Office

PROFESSIONAL EXPERIENCE

Assistant Research Technologist | Arizona State University, Tempe, AZ Dec 2025 – Present

- Own the full design and simulation workflow for a dual-inlet aerosol nozzle used in fuel atomization research, from CAD modeling through CFD validation
- Reduced simulation solve time by 40% through optimized meshing strategy while maintaining accuracy in critical flow regions
- Resolved 8 solver divergence issues through systematic troubleshooting of boundary conditions and initialization parameters
- Delivered 18 post-processing packages including velocity contours and pressure fields to support research documentation and PI review

Design Engineering Intern | Hindustan Aeronautics Limited, Aircraft R&D Center, Bengaluru, India Mar – May 2023

- Led conceptual design of 90-seat regional turboprop aircraft, determining wing area (66.2 m²), MTOW (38,540 kg), and engine selection through constraint analysis per FAR-25
- Designed 3D wing structure in CATIA V5 with 15-rib dual-spar configuration, validating via FEA with 0.84 margin of safety at 2.7% MTOW (1,060 kg)
- Created detailed 2D/3D CAD data packages with GD&T specifications for design reviews and manufacturing handoff
- Co-authored peer-reviewed publication on aircraft design point estimation methodology (Jun 2023)

Manufacturing Engineering Intern | Hindustan Aeronautics Limited, Helicopter Division, Bengaluru, India Mar – Apr 2023

- Supported final assembly of 3 helicopters (2 ALH, 1 LCH) during year-end production ramp, rotating through 7 assembly stations including fuselage integration and systems routing
- Conducted dimensional verification on critical joints with tolerances down to ±0.5mm, identifying a 4mm landing gear alignment issue before downstream assembly
- Participated in root cause analysis when composite panels failed incoming inspection, documenting findings and coordinating corrective action with supplier
- Tracked assembly progress in daily production meetings, helping compile time-study data for process improvement initiatives

PROJECTS

Microchannel Cold Plate Thermal Analysis | ANSYS Fluent, SolidWorks

- Compared serpentine vs manifolded cooling architectures for electronics thermal management, achieving 4.3× lower pressure drop with the manifolded design while maintaining thermal performance

Flat-Fan Spray Nozzle Design | SolidWorks, ANSYS CFD, 3D Printing

- Designed two nozzle variants with different spray angles, validated flow patterns through CFD simulation, and 3D-printed functional prototypes to verify real-world spray distribution

Vortex Shedding CFD Study | ANSYS Fluent 2024 R2

- Analyzed how cylinder cross-section geometry affects oscillating lift forces, finding that elliptical orientation can reduce vibration loads by up to 70% compared to circular

EDUCATION

M.S., Aerospace Engineering Dec 2025
Arizona State University, Tempe, AZ GPA: 3.53/4.0

B.E., Aeronautical Engineering Jun 2023
Visvesvaraya Technological University (VTU), India GPA: 3.96/4.0